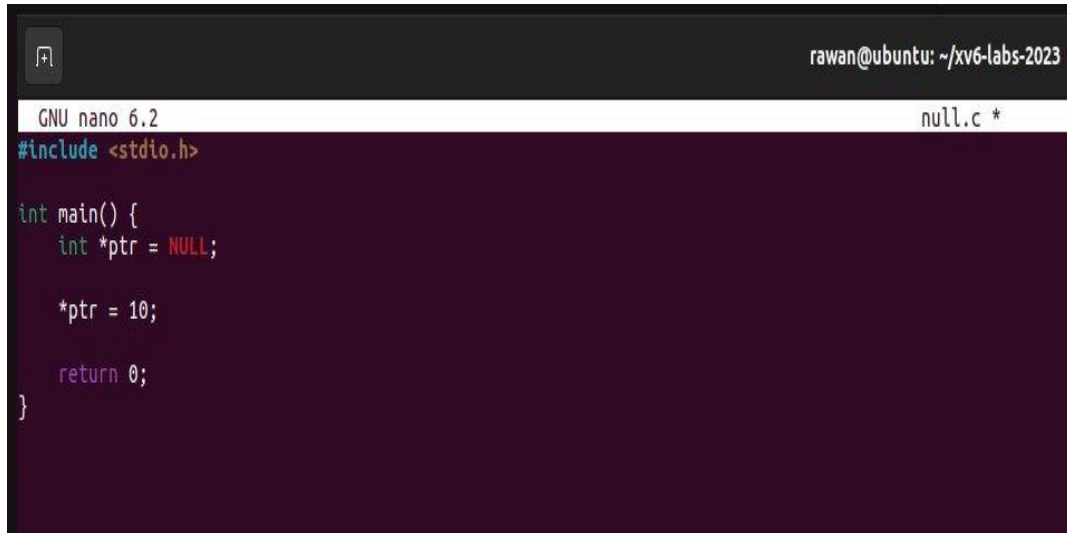


Implementation

Task 1 – NULL Pointer Example

This task demonstrates what happens when a NULL pointer is used incorrectly.



```
rawan@ubuntu: ~/xv6-labs-2023
GNU nano 6.2 null.c *
#include <stdio.h>

int main() {
    int *ptr = NULL;

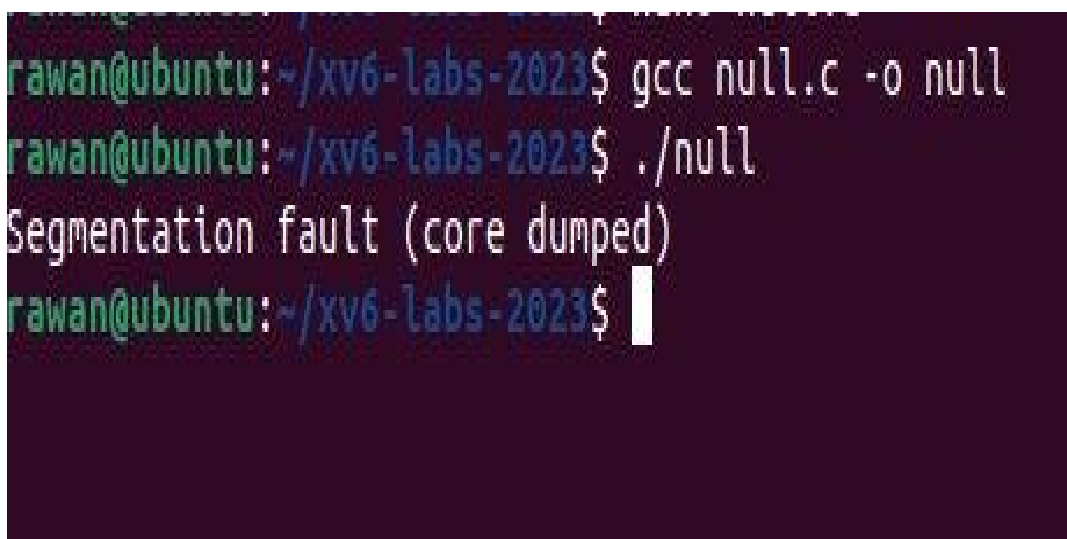
    *ptr = 10;

    return 0;
}
```

The program crashes with a segmentation fault because NULL points to an invalid memory location protected by the OS.

Task 2 – Memory Checking using Valgrind

A program allocates memory then writes outside the valid range.

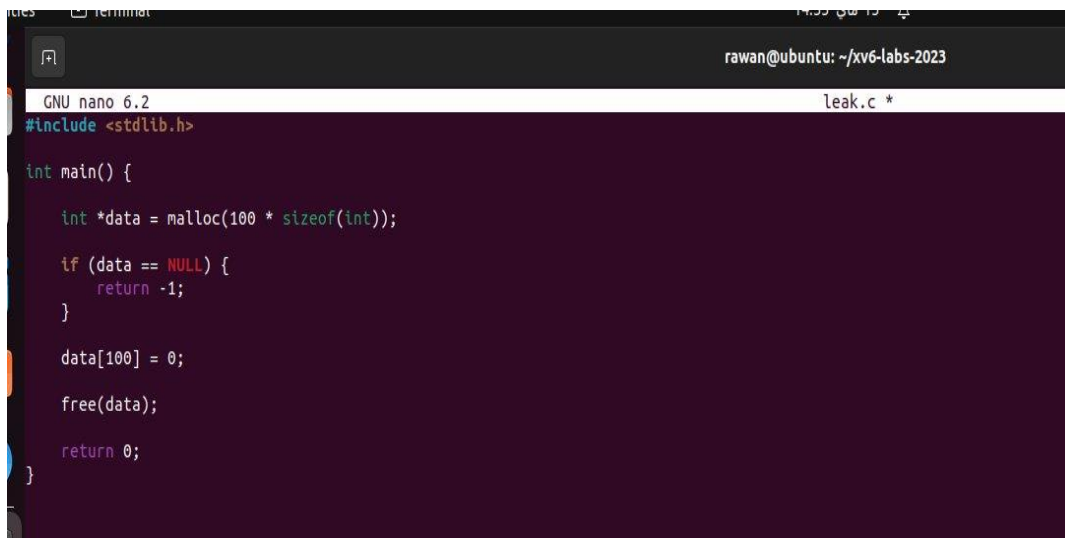


```
rawan@ubuntu:~/xv6-labs-2023$ gcc null.c -o null
rawan@ubuntu:~/xv6-labs-2023$ ./null
Segmentation fault (core dumped)
rawan@ubuntu:~/xv6-labs-2023$
```

Even if the program runs, this is unsafe. Valgrind detects the invalid memory write.

Task 3 – Out-of-Bounds Access

Writing outside array bounds may not crash immediately but leads to undefined behavior.



```
GNU nano 6.2 leak.c *
#include <stdlib.h>

int main() {
    int *data = malloc(100 * sizeof(int));

    if (data == NULL) {
        return -1;
    }

    data[100] = 0;

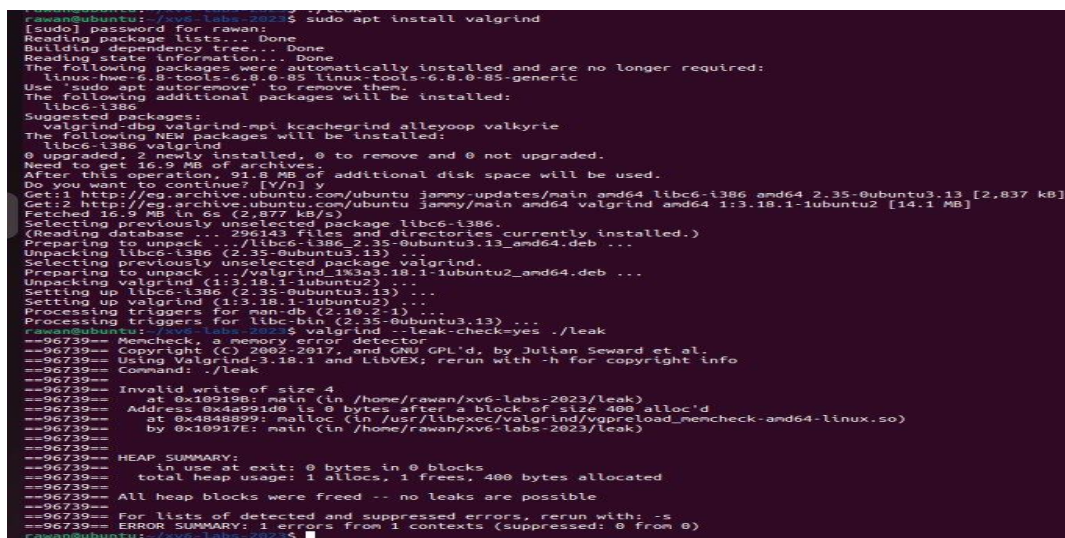
    free(data);

    return 0;
}
```

Tools like Valgrind help identify such hidden errors.

Task 4 – Memory Usage Monitoring

A long-running process allows monitoring memory usage using system tools like ps.



```
rawan@ubuntu: ~/xv6-labs-2023$ sudo apt install valgrind
[sudo] password for rawan:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  linux-hwc-6.8-tools-6.8.0-85 linux-tools-6.8.0-85-generic
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  libc6-i386
Suggested packages:
  valgrind-dbg valgrind-mpi kcachegrind alieyoop valkyrie
The following NEW packages will be installed:
  libc6-i386 valgrind
0 upgraded, 2 newly installed, 0 to remove and 0 not upgraded.
Need to get 16.9 MB of archives.
After this operation, 91.8 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://eg.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libc6-i386 amd64 2.35-0ubuntu3.13 [2,837 kB]
Get:2 http://eg.archive.ubuntu.com/ubuntu jammy/main amd64 valgrind amd64 1:3.18.1-1ubuntu2 [14.1 MB]
Fetched 16.9 MB in 6s (2,877 kB/s)
Selecting previously unselected package libc6-i386.
(Reading database ... 296143 files and directories currently installed.)
Preparing to unpack .../libc6-i386_2.35-0ubuntu3.13_amd64.deb ...
Unpacking libc6-i386 (2.35-0ubuntu3.13) ...
Selecting previously unselected package valgrind.
Preparing to unpack .../valgrind_1:3.18.1-1ubuntu2_amd64.deb ...
Unpacking valgrind (1:3.18.1-1ubuntu2) ...
Setting up libc6-i386 (2.35-0ubuntu3.13) ...
Setting up valgrind (1:3.18.1-1ubuntu2) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for libc-bin (2.35-0ubuntu3.13) ...
rawan@ubuntu: ~/xv6-labs-2023$ valgrind --leak-check=yes ./leak
==96739== Memcheck, a memory error detector
==96739== Copyright (C) 2002-2017, and GNU GPL'd, by Julian Seward et al.
==96739== Using Valgrind 3.18.1 and LibVEX; rerun with -h for copyright info
==96739== Command: ./leak
==96739==
==96739== Invalid write of size 4
==96739== at 0x10919B: main (in /home/rawan/xv6-labs-2023/leak)
==96739== Address 0x4a991d0 is 0 bytes after a block of size 400 alloc'd
==96739==    by 0x10917E: main (in /home/rawan/xv6-labs-2023/leak)
==96739==
==96739== HEAP SUMMARY:
==96739==   in use at exit: 0 bytes in 0 blocks
==96739==   total heap usage: 1 allocs, 1 frees, 400 bytes allocated
==96739==
==96739== All heap blocks were freed -- no leaks are possible
==96739==
==96739== For lists of detected and suppressed errors, rerun with: -s
==96739== ERROR SUMMARY: 1 errors from 1 contexts (suppressed: 0 from 0)
rawan@ubuntu: ~/xv6-labs-2023$
```

VSZ represents virtual memory, while RSS represents actual RAM usage.